

**1. A user reports their laptop won't power on even when plugged in.
List 5 troubleshooting steps you would take to diagnose the cause, and
for each step, explain what you are checking for.**

Step	Troubleshooting Action	What You Are Checking
1. Check the power outlet	Plug another working device into the same outlet.	Confirms whether the wall outlet is supplying power.
2. Check the charger/adapter	Inspect the charger, cable, and connector for damage.	Checks whether the laptop is receiving power from the charger.
3. Check the charging indicator	Connect the charger and look for the charging LED.	If there is no light, the charger, charging port, or power circuit may have a problem.
4. Perform a power reset	Disconnect the charger, remove the battery if removable, hold the power button for about 15–20 seconds, then reconnect power and try again.	Clears residual electrical charge and can resolve a temporary power-state problem.
5. Check internal hardware	If it still does not power on, inspect/reseat accessible RAM and check internal power connections.	Checks for loose RAM or internal connections that may prevent startup.

2. You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue.

Blank Monitor Screen – Troubleshooting Flow

Problem: Monitor is blank, but the PC/CPU appears to be running.

1. Check the monitor power

- Make sure the monitor is switched on and the power LED is lit.
- Check the power cable and wall socket.

2. Check the display cable

- Make sure the HDMI/DisplayPort/VGA cable is firmly connected at both ends.
- Try another cable or another monitor port if available.

3. Check the correct input source

- Use the monitor's **Input/Source** button.
- Select the port actually being used, such as HDMI or DisplayPort.

4. Check the graphics connection

- If the PC has a dedicated GPU, make sure the monitor cable is connected to the **GPU's display port**, not the motherboard port.
- Power off the PC before reseating the GPU if necessary.

5. Check RAM and internal connections

- If there is still no display, power off the PC and reseat the RAM.

- Check for loose GPU power or motherboard connections.

6. Check basic display settings

- If Windows starts but the screen remains blank, try **Win + Ctrl + Shift + B** to reset the graphics driver.
- If possible, connect another monitor to determine whether the problem is with the monitor or PC.

7. Confirm the fix

- Restart the PC and verify that the monitor displays the BIOS/Windows screen normally.
- Test the system for a few minutes to ensure the blank-screen problem does not return.

3. A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.

Wireless Mouse Not Working – Troubleshooting Steps

Since the keyboard works, the PC is powered on and responding. I would isolate the problem in this order:

1. Check the mouse itself

- Make sure the mouse power switch is ON and the battery is charged/replaced.
- Check whether the mouse's LED/sensor light responds.
- If possible, test the mouse on another computer.
- **If it doesn't work there → the mouse is likely faulty.**

2. Check the wireless receiver/USB port

- Remove the mouse's USB receiver and plug it into another USB port.
- Try a different known-working USB device in the original port.
- **If another device also fails → the USB port may be faulty.**

3. Check the wireless connection

- If the mouse has a pairing/connect button, pair it again with its receiver.
- Keep the mouse close to the receiver and remove possible wireless interference.

4. Check system settings

- Open **Settings** → **Bluetooth & devices** → **Mouse** and confirm the mouse is detected.
- In **Device Manager**, check **Mice and other pointing devices** and **Human Interface Devices** for errors.
- Update or reinstall the mouse/USB driver if necessary.

5. Restart and confirm

- Restart the computer and test the mouse again.
- If the mouse works after changing the USB port, the original port was likely the problem.
- If it works on another PC but not this one, the issue is more likely with the computer's USB, drivers, or system setting

4. Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one.

Hardware Cause	How to Check
1. Faulty or slow cooling fan	Check whether the CPU/GPU/case fans are spinning properly. Listen for unusual noise and make sure no fan is stuck. Replace a faulty fan if needed.
2. Dust buildup	Turn off and unplug the PC, then inspect the fans, heatsinks, and air vents for dust. Clean them carefully with compressed air and make sure airflow is not blocked.
3. Old/dried thermal paste	If the CPU temperature remains high after cleaning and the fan works normally, inspect the CPU cooler. Remove the cooler, clean the old thermal paste, apply fresh thermal paste, and reinstall the cooler properly.